NOAA FISHERIES

Economic and Social Analysis

FishSET is a decision-support toolbox for better and more timely analyses to understand how fishing choices are affected by changing conditions.

FishSET: A Spatial Economics Toolbox for Better Fisheries Management

FishSET is a set of statistical programming and data management tools, developed as an R package to improve fishery modeling. The tools standardize fisheries data management, enable the use of location choice models to provide input into fishery management, and provide various modeling and visualization tools.

Overcoming Challenges

Predictive models of fishing behavior are valuable tools for sustainable fisheries management and conservation. FishSET enables the development of these models for specific use cases by:

- **Streamlining Data:** It simplifies the difficult tasks of preparing, combining, and updating diverse data sources.
- **Guiding Model Selection:** It assists users in choosing the most appropriate predictive models.
- **Clarifying Results:** It helps in the interpretation of complex model outputs.

FishSET combines multiple types of tools, approaches, and information into one place. This allows for easier analysis of how different factors might predict fisher behavior. It can combine biological and economic data to identify policy options that help achieve conservation and socioeconomic goals of science-based fisheries management.

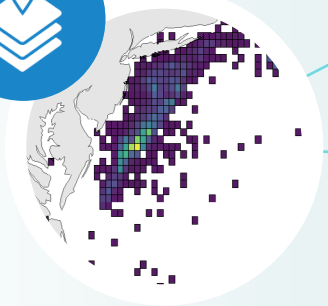
Benefits of the Toolbox

- **Superior Integration:** Provides advanced data organization, analysis, and integration for spatial models.
- **Best Practices:** Supports best practices for data management, modeling, and model comparison.
- **Ease of Use:** Combines multiple modeling approaches in a single toolbox for ease of model comparison and use. Includes an optional Graphical User Interface for less technical users.

Artwork (Top): Bob Hines

Types of Tools

DATA TOOLS



Data Management and Integration

Facilitates the development and integration of fisheries datasets for spatial modeling.

Data Analysis and Mapping

Creates graphs, allows the visualization of geographic data, and prepares data for spatial modeling.

MODELING TOOLS

Model Design and Selection

Enables the creation and testing of different combinations of variables and model specifications.

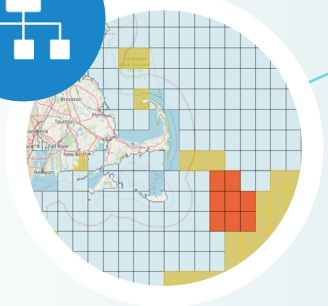
Modeling

Runs a variety of spatial discrete choice models, including standard, cutting-edge, and user-designed options.

Model Comparison and Reporting

Offers an extensive set of model performance evaluations and provides summaries of data, models, and results.

POLICY TOOLS



Policy Simulation

Estimates the potential impacts on fisheries from various scenarios, including fishing ground closures resulting from regulations; new ocean uses, such as Marine Protected Areas or offshore energy projects; and significant events, such as hurricanes or oil spills.

Learn More

To learn more and get started with FishSET, visit the website:

https://noaa-nwfsc.github.io/FishSET/

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